Vrishab Prasanth Davey

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EDUCATION

University of Ottawa — **Ottawa, Ontario**
MCS and Concentration Applied Artificial Intelligence
- Machine Learning, NLP, Virtual Assistants, Ethics AI, Software Engineering

SRM Institute of Science and Technology — **Chennai, India**
BTech in Computer Science and Engineering(GPA: 3.8)
- Excellence awardee, Overall 2nd place in FB'22, AIR 50 - NEO, Research Assistantship

WORK EXPERIENCE

SRM Institute of Science & Technology — **Chennai, India**
Researcher — *Dec 2023 - May 2024*
- **Developed** a lightweight **U-Net model** for **brain tumor segmentation** on the **BITE MRI dataset**, achieving **84% mean IoU** across **846 images**, surpassing traditional methods without data augmentation for faster, reliable clinical use.
- **Enhanced model robustnes**s by training on **coronal, sagittal, and transversal MRI planes**, increasing **mean IoU by 5%** across views, **with 99%** pixel accuracy on the transversal plane, ensuring **multi-perspective diagnostic reliability**.
- **Benchmarked** U-Net against **K-Means, Fuzzy C-Means, and LinkNet**, achieving a **4% higher** mean IoU (**up to 89%**) and **greater pixel accuracy**, highlighting its real-time clinical segmentation capabilities.

Wipro — **Chennai, India**
Software Developer — *Jan 2024 - Mar 2024*
- **Designed** and **deployed** a **Virtual Agent** on **ServiceNow**, integrating **ITSM Virtual Agent** for enhanced response times, achieving a **30%** improvement.
- **Configured 20+ conversation flows** in **Virtual Agent Designe**r and **optimized Natural Language Understanding (NLU)** for **95% accuracy** in intent recognition.
- **Integrated OpenAI API** to handle complex queries, **fine-tuned** a custom **Retrieval-Augmented Generation (RAG)** model, achieving **90%+ accuracy** in **document-based responses** and significantly **reducing manual lookup time.**
- **Deployed solutions** as **Flask applications**, leading to a **40% increase** in **self-service resolutions.**

Samsung R&D Institute India — **Bangalore, India**
Machine Learning Engineer — *Mar 2023 - Nov 2023*
- **Spearheaded** the **development** of a **warping-based CNN** for **real-time Eye Gaze Correction** on live video, **achieving 33 FPS** on consumer laptops, **addressing parallax issues**, and improving eye contact in video calls.
- **Enhanced** gaze correction accuracy by implementing **shape- and color-based loss functions**, reducing **sclera** and **eyelid artifacts** by **20%**, and in**creasing PSNR** by **4 dB** over models like Deep Learning's **DeepWarp**.
- **Validated** model on **diverse datasets**, including **DIRL** and **CAVE**, achieving **84.5% realism recognitio**n across **ethnicities** and **consistent gaze correction** within a **±30-degree range**.
- **Optimized** model with **architectural refinements** in **TensorFlow**, reducing **computation time** by **20%** and enabling **seamless gaze correction** in **live video streaming**.

PROJECTS

1. U-Net for MRI Brain Tumor Segmentation
Developed a U-Net model for automated MRI brain tumor segmentation, **achieving 93.86% accuracy** and a **high Dice coefficient.** Integrated a Streamlit interface for real-time visualization, enhancing accessibility in **neuro-oncology** workflows.

2. Eye Gaze Correction for Video Conferencing
Constructed a **real-time Eye Gaze Correction system** with a **warping-based CNN**, achieving **33 FPS** and reducing eye contact issues by **minimizing parallax.** Custom **loss functions reduced visual artifacts** by **20%**, validated across datasets with an **84.5% realism recognition rate**.

3. Streamlining Software Ticket Handling: A Case Study with Transformers and Generative Models
Engineered a **robust automated ticket triaging system** integrated with **Jira**, leveraging fine-tuned **transformer/LLM models (BERT)** and **GPT-4o for classification, prioritization, and team assignment tasks**. Achieved a **90.2% F1-score** in ticket classification and an **87% accuracy** using role-based prompt engineering techniques, optimizing domain-specific NLP workflows. Implemented semantic similarity with **Sentence Transformers** for dynamic team assignments, streamlining software project management by reducing manual errors and accelerating response times in fast-paced development environments.

SKILLS & INTERESTS

Programming Languages: Python, MySQL, PostgreSQL, C, C++, TensorFlow, Keras, Sklearn, MLOps, NLTK, AWS, Flask, BeautifulSoup, Nodejs, CSS, Git, Docker, Apache Hadoop, HTML, Excel, Power BI, MS Excel, MS Word.

Certifications: Machine Learning Specialization, Introduction to Data Science Generative, AWS Academy Cloud Foundations, Oracle Database Foundations Certified Junior Associate, Database Management Essentials.

Publications: 1. Vrishab Davey, Dr. Akilandeswari P. "**Streamlined U-Net for MRI Brain Tumor Segmentation with Streamlit Visualization Interface." IC2E3, IEEE, 2024**. **2. Vrishab Davey**, Dr. Akilandeswari P. **"Eye Contact During Live Video Conferencing." IC2E3, IEEE, 2024**.